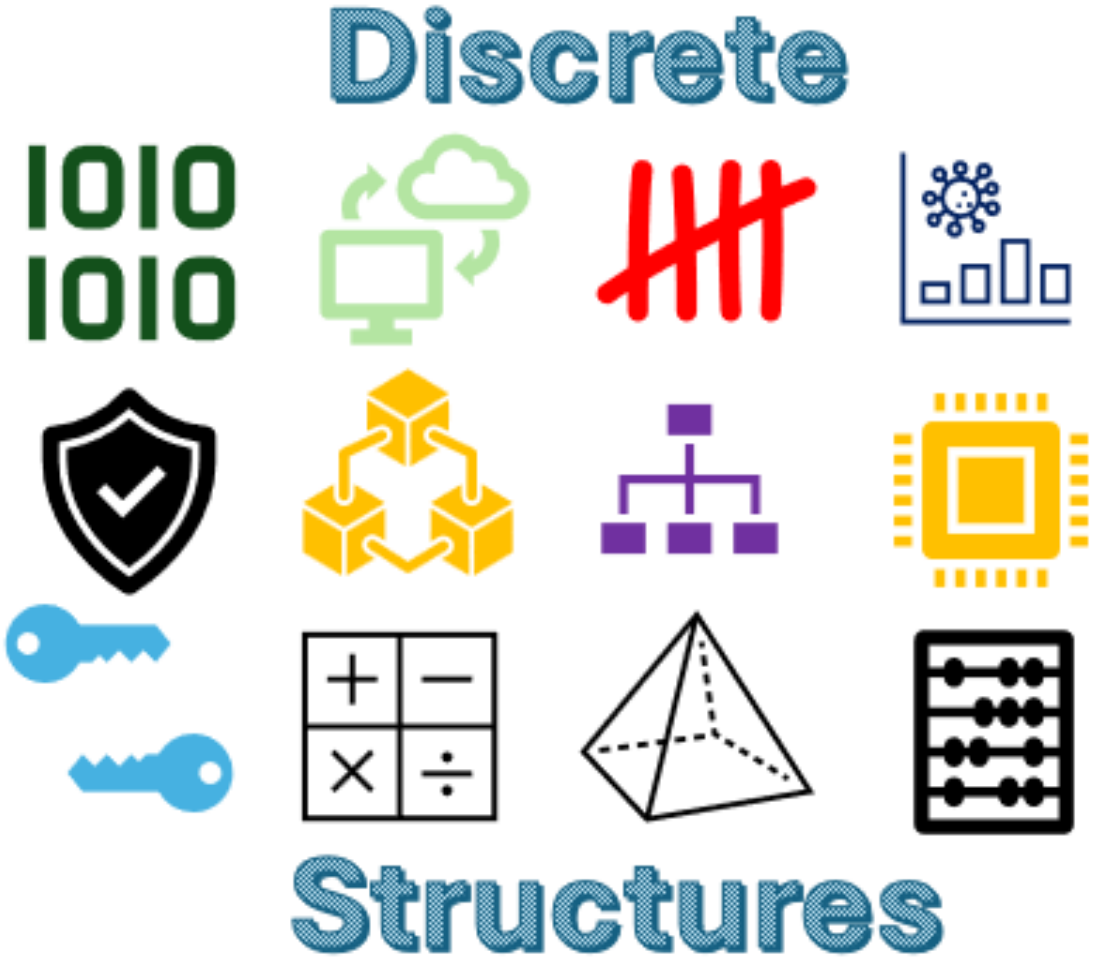
COMP 2310: Discrete Structures
Fall 2026

Chapter 4: Relation and digraph



4.6. Data structures for Relations and Diagraph

Data structures for Relations

- Data can be stored in a list known as an array.
- The locations of an array are consecutively numbered in a computer memory.

Data structures for Relations

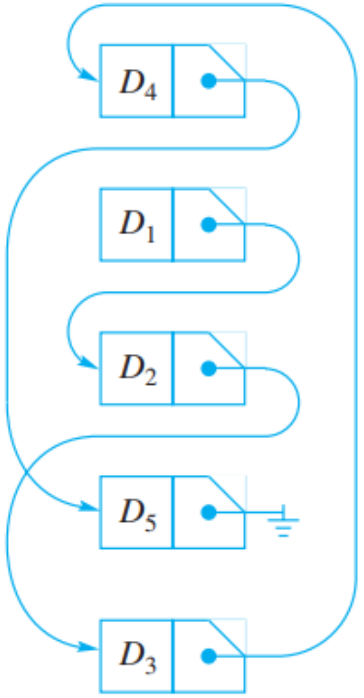
- Array A with 5 data items

Index Location		
A[1]	D_1	Data 1
A[2]	D_2	Data 2
A[3]	D_3	Data 3
A[4]	D_4	Data 4
A[5]	D_5	Data 5

With array, it would be difficult to add an item between D_2 and D_4

Data structures for Relations

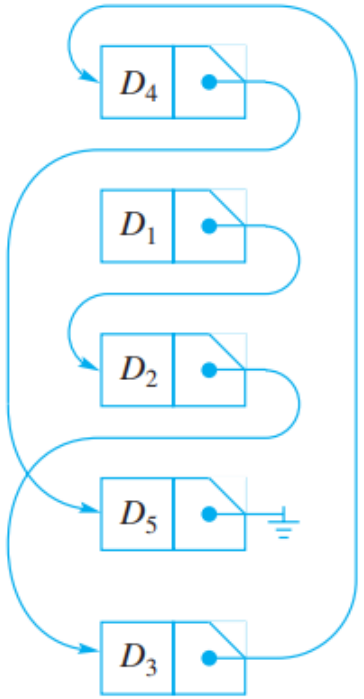
- Using **linked list** to store data
- Consists of two linear arrays: **Data** array **A** and a **Pointer** array **P**
- Physical order does not have to be the same as the logical order



	A	P
	D_4	4
	D_1	3
	D_2	5
	D_5	0
	D_3	1

Data structures for Relations

- Insert a new data between D2 and D3



A
D_4
D_1
D_2
D_5
D_3

P
4
3
5
0
1

Previous representation

START 2

A
D_4
D_1
D_2
D_5
D_3
E

P
4
3
6
0
1
5

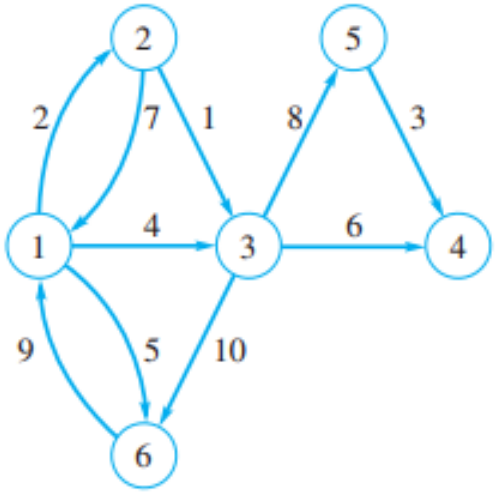
New representation, Start at 2 since D1 is located at 2

Data structures for Relations

- Insert a new data between D4 and D5, what are the new A and P?

Data structures for Relations and Diagraphs

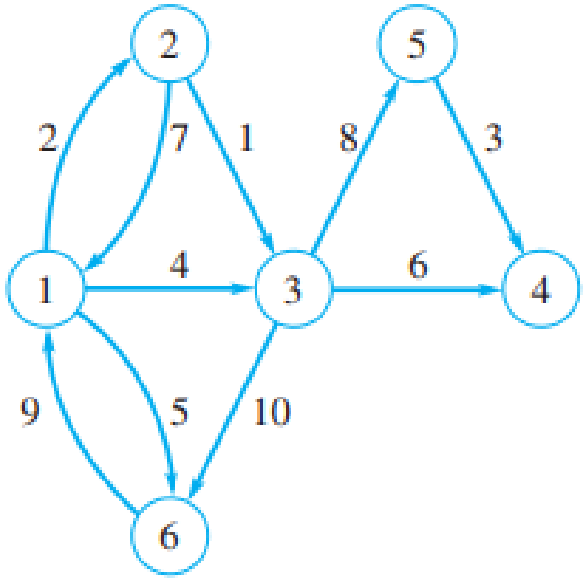
- Consider the following graph:



- We see that we have 10 edges. We can conclude that we have 10 pairs.
- R = {(1,2),(1,6),(1,3),(2,1),(2,3),(3,5),(3,4),(3,6),(5,4),(6,1)}

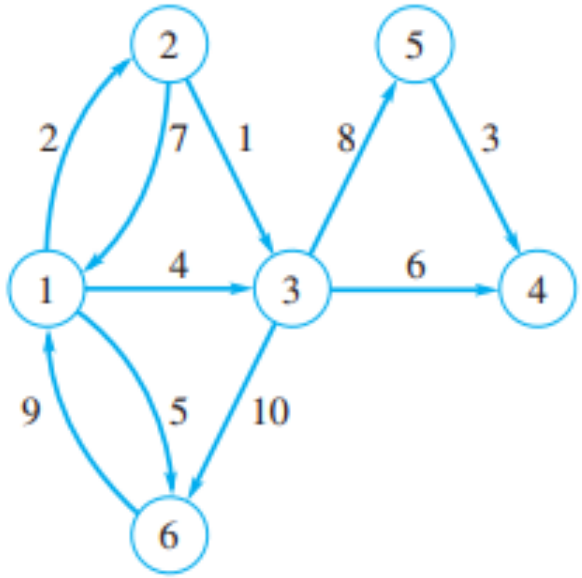
Terms for representing a digraph into linked list

- **VERT**
  - Points to the first outgoing edge for each vertex.
- **TAIL**
  - Beginning vertex
- **HEAD**
  - Ending vertex
- **NEXT**
  - Points to the next outgoing edge in the linked list for a given vertex.



Representing a digraph into linked list

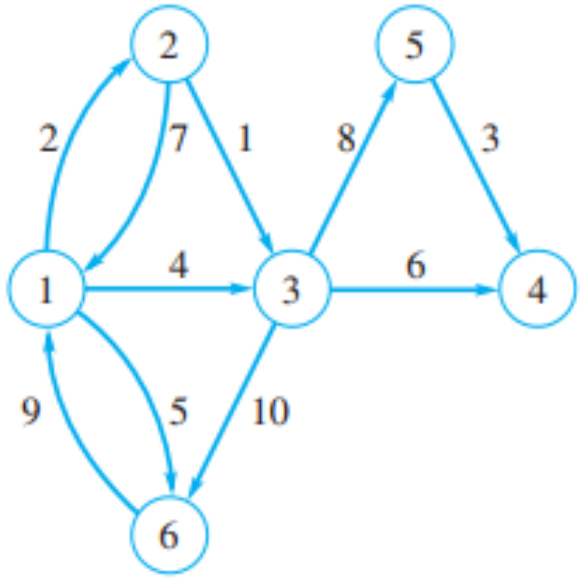
- 0 in next indicates that no other edges left to traverse for the starting vertex



VERT	TAIL	HEAD	NEXT
10	1	2	0
2	2	3	3
4	2	1	0
0	3	5	6
5	5	4	0
8	3	4	7
	3	6	0
	6	1	0
	1	6	1
	1	3	9

Representing a digraph into linked list

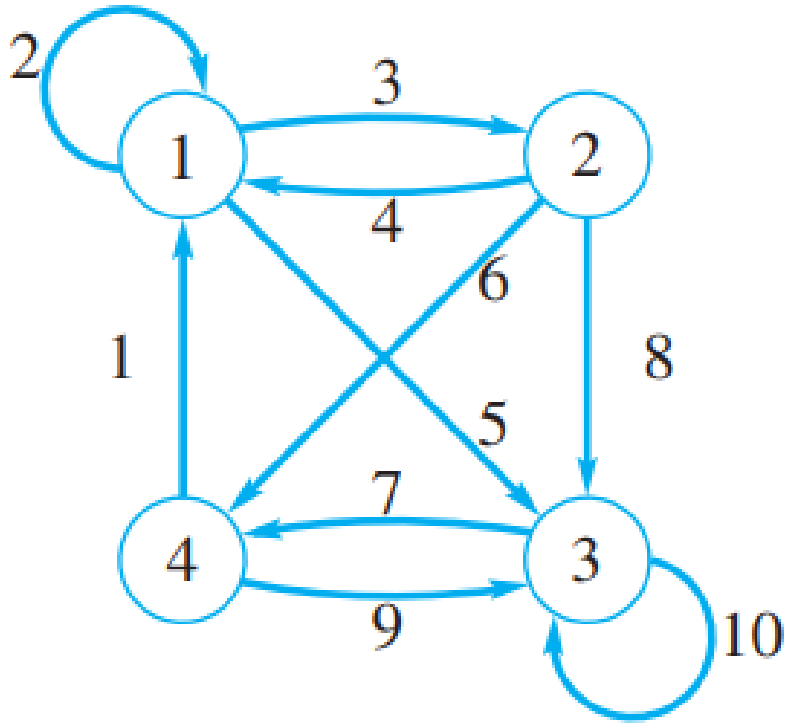
- Alternative representation of the diagraph than the previous one



VERT	TAIL	HEAD	NEXT
9	1	2	0
3	2	3	0
6	2	1	2
0	3	5	7
5	5	4	0
8	3	4	4
	3	6	0
	6	1	0
	1	6	10
	1	3	1

Example

- Let $A = \{1, 2, 3, 4\}$ and let R be the relation whose digraph is shown in Figure 4.37. Describe arrays VERT, TAIL, HEAD, and NEXT, setting up a linked-list representation of R, so that the edges out of each vertex are reached in the list in increasing order.



Operations on relations: Complementary relation

- Let A = {1, 2, 3, 4} and B = {a, b, c}. Let

  R = {(1,a), (1, b), (2, b), (2, c), (3, b), (4,a)}

  S = {(1, b), (2, c), (3, b), (4, b)}.

- **Compute**
  - $\bar{R}$ (Complementary relation)
    - The complement of the relation R (denoted as $\bar{R}$) is defined as the set of all pairs in A x B that are not in R
      - $\bar{R}$ **= A x B – R**

        = {(1, a), (1, b), (1, c), (2, a), (2, b), (2, c), (3, a), (3, b), (3, c), (4, a), (4, b), (4, c)} - ***{(1,a), (1, b), (2, b), (2, c), (3, b), (4,a)}***

        = {(1, c), (2, a), (3, a), (3, c), (4, b), (4, c)}

Operations on relations: Inverse relation

- Let A = {1, 2, 3, 4} and B = {a, b, c}. Let

  $R = \{(1,a), (1, b), (2, b), (2, c), (3, b), (4,a)\}$

  $S = \{(1, b), (2, c), (3, b), (4, b)\}$.

- **Compute**
  - R^{-1} (Inverse relation)
  - $R^{-1} = \{(a, 1), (b, 1), (b, 2), (c, 2), (b, 3), (a, 4)\}$